

Welcome to the minitutorial!

**To encourage interaction, please come to the ground floor
and site near the front row.**

<https://github.com/PerformanceEstimation/Tutorial-SIAM-OP26>



Performance Estimation Problems (PEPs)

Systematic analysis and design of first-order optimization algorithms
via the performance estimation framework

François Glineur, Adrien Taylor



SIAM Conference on Optimization, tutorial session – June 2026



PEPs on an example: gradient descent

PEPs and proof structures

PEPs to design a (surprising?) algorithm

PEPs on an example: gradient descent

PEPs and proof structures

PEPs to design a (surprising?) algorithm

| Concrete example: analysis of a gradient step

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ (continuously differentiable). Find $x_\star \in \mathbb{R}^d$ such that

$$f(x_\star) = \min_{x \in \mathbb{R}^d} f(x),$$

with f is L -smooth and μ -strongly convex ($f \in \mathcal{F}_{\mu,L}$).

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Question: what *a priori* guarantees after N iterations?

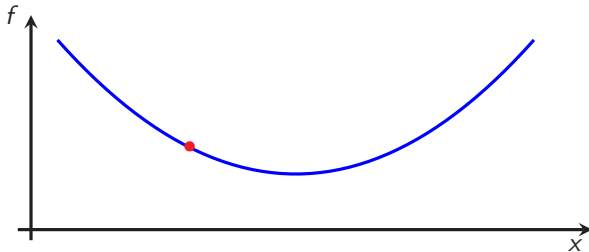
Examples: what about $f(x_N) - f(x_\star)$, $\|\nabla f(x_N)\|$, $\|x_N - x_\star\|$?

| About the assumptions

A differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is μ -strongly convex and L -smooth iff $\forall x, y \in \mathbb{R}^d$:

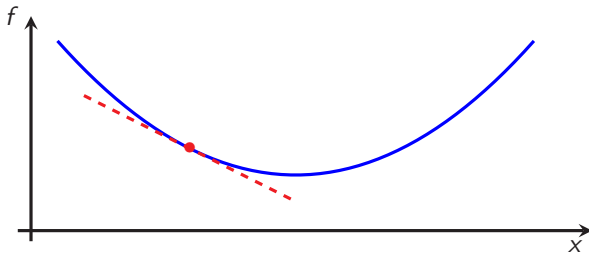
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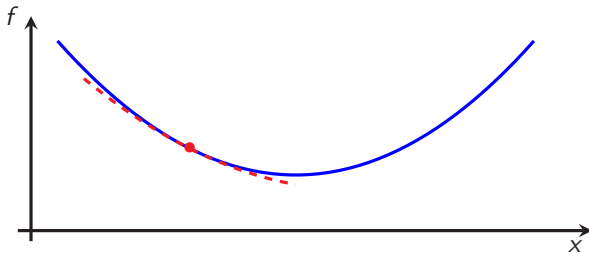
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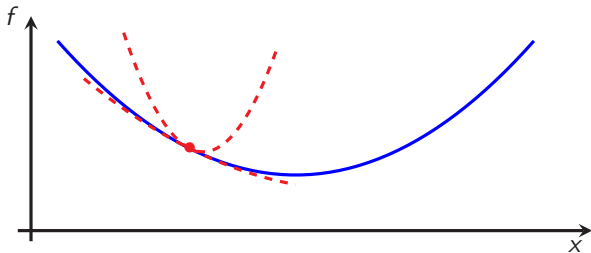


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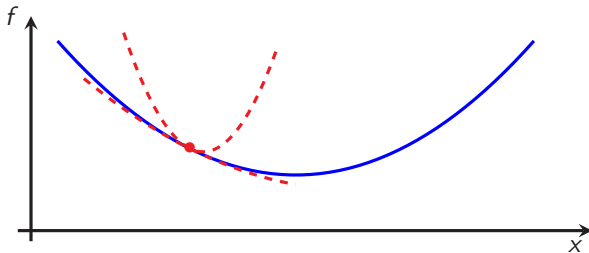
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(1&2) $\langle \nabla f(x) - \nabla f(y); x - y \rangle \geq \frac{1}{L+\mu} \|\nabla f(x) - \nabla f(y)\|^2 + \frac{\mu L}{L+\mu} \|x - y\|^2$.

| Convergence rate of a gradient step

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Toy example: What is the smallest τ such that:

$$\|x_1 - x_\star\|^2 \leq \tau \|x_0 - x_\star\|^2,$$

for all

- ◇ $d \in \mathbb{N}$, L -smooth and μ -strongly convex function f (notation $f \in \mathcal{F}_{\mu,L}$),
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Computing τ ?¹

$$\tau = \max_{f, x_0, x_1, x_\star} \frac{\|x_1 - x_\star\|^2}{\|x_0 - x_\star\|^2}$$

s.t. $f \in \mathcal{F}_{\mu,L}$

Functional class

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Variables: $f, x_0, x_1, x_\star$.

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Algorithm

Parameters: μ, L, γ .

$$\nabla f(x_\star) = 0$$

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- ◇ replacing them by

$$\begin{aligned} f_\star &\geq f_0 + \langle g_0, x_\star - x_0 \rangle + \frac{1}{2L} \|g_\star - g_0\|^2 + \frac{\mu}{2(1-\mu/L)} \|x_\star - x_0 - \frac{1}{L}(g_\star - g_0)\|^2 \\ f_0 &\geq f_\star + \langle g_\star, x_0 - x_\star \rangle + \frac{1}{2L} \|g_0 - g_\star\|^2 + \frac{\mu}{2(1-\mu/L)} \|x_0 - x_\star - \frac{1}{L}(g_0 - g_\star)\|^2. \end{aligned}$$

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- ◇ Same optimal value (no relaxation): **non-convex quadratic** problem.

| Semidefinite lifting

◇ Define $P \triangleq [x_0 - x_*, g_0] \in \mathbb{R}^{d \times 2}$ and $F \triangleq f_0 - f_*$.

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(using homogeneity argument and substituting x_1 and g_*).

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◇ Assuming $x_0, x_*, g_0 \in \mathbb{R}^d$ with $d \geq 2$, same optimal value as original problem!

| Interactive parts begin!



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| Numerical worst-case bound from the SDP

Set $L = 1$, $\mu = .1$ and solve SDP for a few γ 's.



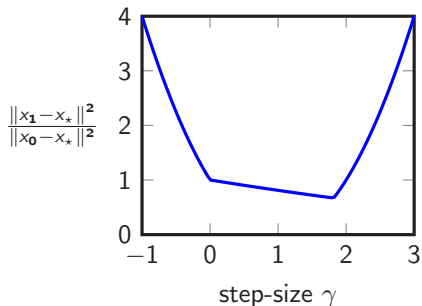
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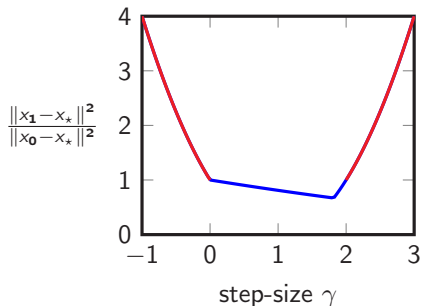


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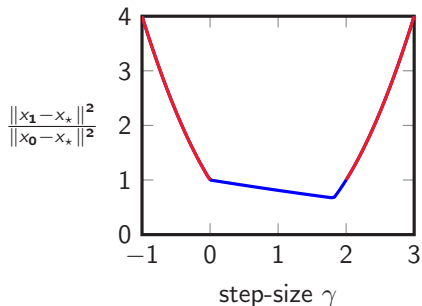
Can you guess the closed form?

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Can you guess the closed form? Observation: numerics match the known $\max\{(1 - \gamma L)^2, (1 - \gamma \mu)^2\}$.

| Dual problem

◇ Primal problem (abstract form)

$$\begin{aligned} \max_{G, F} \quad & \langle b_{\text{obj}}, F \rangle + \text{Tr}(A_{\text{obj}} G) \quad \text{subject to:} \quad \langle b_1, F \rangle + \text{Tr}(A_1 G) \leq 0 \\ & \langle b_2, F \rangle + \text{Tr}(A_2 G) \leq 0 \\ & \langle b_{\text{init}}, F \rangle + \text{Tr}(A_{\text{init}} G) = 1 \\ & G \succcurlyeq 0 \end{aligned}$$

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- ◇ Lagrangian relaxation (provides upper bounds for any $\lambda_1, \lambda_2 \geq 0, S \succcurlyeq 0, \tau \in \mathbb{R}$):

$$\max_{G, F} \tau + \langle [b_{\text{obj}} - \lambda_1 b_1 - \lambda_2 b_2 - \tau b_{\text{init}}], F \rangle + \text{Tr}([A_{\text{obj}} - \lambda_1 A_1 - \lambda_2 A_2 - \tau A_{\text{init}} + S] G)$$

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- ◇ Dual problem (minimize upper bound)

$$\begin{aligned} \min_{\substack{\lambda_1, \lambda_2 \geq 0 \\ S \succcurlyeq 0 \\ \tau \in \mathbb{R}}} \tau \quad \text{subject to:} \quad & 0 = b_{\text{obj}} - \lambda_1 b_1 - \lambda_2 b_2 - \tau b_{\text{init}} \\ & S = \lambda_1 A_1 + \lambda_2 A_2 + \tau A_{\text{init}} - A_{\text{obj}} \end{aligned}$$

| Dual solutions (numerical) — back to gradient descent

Set $L = 1$, $\mu = .1$ and recover λ_1, λ_2 for a few γ 's.



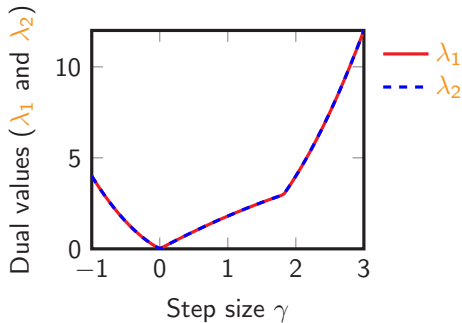
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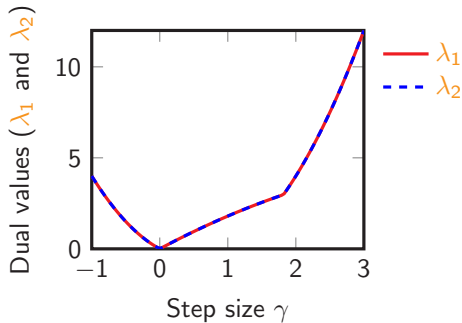


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Numerics match $\lambda_1 = \lambda_2 = 2|\gamma|\rho(\gamma)$ with $\rho(\gamma) = \max\{\gamma L - 1, 1 - \gamma\mu\}$.

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◇ Weak duality: any dual feasible point $\equiv$ valid worst-case convergence rate ($\Uparrow$):

$$\|x_1 - x_\star\|^2 \leq \tau \|x_0 - x_\star\|^2 \text{ for all } f \in \mathcal{F}_{\mu, L}, \text{ all } x_0 \in \mathbb{R}^d, \text{ all } d \in \mathbb{N}, \text{ with}$$
$$x_1 = x_0 - \gamma \nabla f(x_0).$$



$$\exists \lambda, \tau \geq 0 : \begin{bmatrix} \tau - 1 + \frac{\lambda L \mu}{L - \mu} & \gamma - \frac{\lambda (L + \mu)}{2(L - \mu)} \\ \gamma - \frac{\lambda (L + \mu)}{2(L - \mu)} & \frac{\lambda}{L - \mu} - \gamma^2 \end{bmatrix} \succcurlyeq 0$$

| Dual problem — back to gradient descent

- ◇ Dual problem is

$$\min_{\tau, \lambda_1, \lambda_2 \geq 0} \tau \quad \text{subject to} \quad S = \begin{bmatrix} \tau - 1 + \frac{\lambda_1 L \mu}{L - \mu} & \gamma - \frac{\lambda_1 (L + \mu)}{2(L - \mu)} \\ \gamma - \frac{\lambda_1 (L + \mu)}{2(L - \mu)} & \frac{\lambda_1}{L - \mu} - \gamma^2 \end{bmatrix} \succcurlyeq 0$$
$$0 = \lambda_1 - \lambda_2.$$

- ◇ Weak duality: any dual feasible point $\equiv$ valid worst-case convergence rate ($\Uparrow$):

$$\|x_1 - x_\star\|^2 \leq \tau \|x_0 - x_\star\|^2 \text{ for all } f \in \mathcal{F}_{\mu, L}, \text{ all } x_0 \in \mathbb{R}^d, \text{ all } d \in \mathbb{N}, \text{ with}$$
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- ◇ Strong duality holds ($\exists$ Slater point): any valid worst-case convergence rate $\equiv$ valid dual feasible point ($\Downarrow$).

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| Dual certificate for gradient descent

Guessed expression for the multipliers: does it yield valid proof?



Your turn! Complete and run the corresponding notebook cells.

| Dual problem, sums-of-squares, and standard proofs

◇ Primal problem (abstract form)

$$\begin{aligned} \max_{G, F} \langle b_{\text{obj}}, F \rangle + \text{Tr}(A_{\text{obj}} G) \quad \text{subject to:} \quad & \langle b_1, F \rangle + \text{Tr}(A_1 G) \leq 0 \\ & \langle b_2, F \rangle + \text{Tr}(A_2 G) \leq 0 \\ & \langle b_{\text{init}}, F \rangle + \text{Tr}(A_{\text{init}} G) = 1 \\ & G \succcurlyeq 0 \end{aligned}$$

| Dual problem, sums-of-squares, and standard proofs

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◇ Dual problem (minimize upper bound)

$$\begin{aligned} \min_{\substack{\lambda_1, \lambda_2 \geq 0 \\ S \succcurlyeq 0 \\ \tau \in \mathbb{R}}} \tau \quad \text{subject to:} \quad & 0 = b_{\text{obj}} - \lambda_1 b_1 - \lambda_2 b_2 - \tau b_{\text{init}} \\ & S = \lambda_1 A_1 + \lambda_2 A_2 + \tau A_{\text{init}} - A_{\text{obj}} \end{aligned}$$

| Dual problem, sums-of-squares, and standard proofs

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| Dual problem, sums-of-squares, and standard proofs

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| Dual problem, sums-of-squares, and standard proofs

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| Dual problem, sums-of-squares, and standard proofs

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| Dual problem, sums-of-squares, and standard proofs

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| Dual problem, sums-of-squares, and standard proofs

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| Dual problem, sums-of-squares, and standard proofs

Gradient with $\gamma = \frac{1}{L}$. Perform weighted sum of two inequalities

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$$f_0 \geq f_\star + \frac{1}{2L} \|\nabla f(x_0)\|^2 + \frac{\mu}{2(1-\mu/L)} \left\| x_0 - x_\star - \frac{1}{L} \nabla f(x_0) \right\|^2 \quad : \lambda_1$$

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$$\|x_1 - x_\star\|^2 \leq (1 - \gamma\mu)^2 \|x_0 - x_\star\|^2 - \underbrace{\gamma \frac{2-\gamma(L+\mu)}{L-\mu} \|\mu(x_0 - x_\star) - \nabla f(x_0)\|^2}_{\text{non-negative}},$$

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leading to $\|x_1 - x_\star\|^2 \leq (1 - \gamma\mu)^2 \|x_0 - x_\star\|^2$

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leading to $\|x_1 - x_\star\|^2 \leq (1 - \gamma\mu)^2 \|x_0 - x_\star\|^2$ (tight).

PEPs on an example: gradient descent

PEPs and proof structures

PEPs to design a (surprising?) algorithm

| Insight into proof structures from PEPs

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ (continuously differentiable) with $f \in \mathcal{F}$. Find $x_\star \in \mathbb{R}^d$ such that

$$f(x_\star) = \min_{x \in \mathbb{R}^d} f(x).$$

For example, assume $(x_\star = 0, f_\star = 0$ for readability):

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For example, assume ($x_\star = 0$, $f_\star = 0$ for readability):

◇ Gradient-based method, e.g.: $x_k = x_{k-1} - \sum_{i=0}^{k-1} \gamma_{k,i} \nabla f(x_i)$,

| Insight into proof structures from PEPs

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ (continuously differentiable) with $f \in \mathcal{F}$. Find $x_\star \in \mathbb{R}^d$ such that

$$f(x_\star) = \min_{x \in \mathbb{R}^d} f(x).$$

For example, assume ($x_\star = 0$, $f_\star = 0$ for readability):

- ◇ Gradient-based method, e.g.: $x_k = x_{k-1} - \sum_{i=0}^{k-1} \gamma_{k,i} \nabla f(x_i)$,
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Then, proofs of the fact “Perf $\leq \tau$ Init” can be framed as: $\exists (\lambda_i)_i, \lambda_i \geq 0$, and $S \succcurlyeq 0$

$$0 \geq \sum_i \lambda_i (\text{Tr}(G A_i) + b_i^T F) = \text{Perf} - \tau \text{Init} + \text{Tr}(G S).$$

| Gradient descent... again!

Toy example: What is the smallest τ such that:

$$f(x_1) - f_{\star} \leq \tau(f(x_0) - f_{\star}),$$

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Wooclap time!

<https://app.wooclap.com/THIKNZU?from=instruction-slide>



PEPs on an example: gradient descent

PEPs and proof structures

PEPs to design a (surprising?) algorithm

| Designing algorithms

A “generic” first-order method

$$w_1 = w_0 - \gamma_{1,0} \nabla f(w_0)$$

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$\vdots$

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(FOM)

for some coefficients $\{\gamma_{i,j}\}$. Generic **non-adaptive** first-order method.

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- ◇ pick a performance criterion, for instance $\frac{\|w_n - w_\star\|^2}{\|w_0 - w_\star\|^2}$,
- ◇ solve the minimax (minimize worst-case): $\min_{\{\gamma_{i,j}\}_{i,j}} \max_{f \in \mathcal{F}, \{w_i\}} \frac{\|w_n - w_\star\|^2}{\|w_0 - w_\star\|^2}.$

| Hands-on

Design an algorithm for minimizing
 L -smooth μ -strongly convex functions.

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Your turn! Complete and run the corresponding notebook cells.

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- ◇ Goal 1: design an optimized two-step gradient method

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$$x_2 = x_1 - \gamma_1 \nabla f(x_1),$$

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| Concluding remarks

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Byproducts:

- ◇ computer-assisted design of analyses,
- ◇ computer-assisted design of numerical methods,
- ◇ step towards reproducible theory: validation & benchmark tool for proofs!